Detecting and Removing Gender Bias in Word Embeddings

Ariyan Hossain, Rakinul Haque, Khondokar Mohammad Ahanaf Hannan, Nowreen Tarannum Rafa, Humayra Musarat

Department of Computer Science and Engineering, Brac University

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Abstract

When misused, machine learning may reveal human-induced data biases. Word embedding, a popular paradigm for collecting textual input as vectors for machine learning and natural language processing, causes this complexity. To that aim, we will present a way for altering an embedding to remove undesired connections like "nurse" and "female" while retaining others like "queen" and "female". We'll concentrate on BERT, a transformer model that generates contextual word embeddings. Gender direction and BERT's fill-in-the-mask task reveal gender prejudice. To debias the embedding, we will remove its component along the gender subspace and ensure that every gender-neutral term is separate from gender-specific word pairings.

Keywords: Natural Language Programming; Bias Detection; Debiasing; Word embeddings; BERT; Gender Bias

Introduction

Word embeddings are word representations that mediate between the human and computer perceptions of words. They are used in NLP to denote the representation of words for text analysis, but can also reveal biases. Recent research suggests that word embeddings may be skewed towards one gender. Word embeddings have been the subject of hundreds of academic studies, but none has emphasized the severity of the sexism they have especially from contextualized word embeddings. We propose a systematic and concise strategy to detect and reduce gender bias in word embeddings.

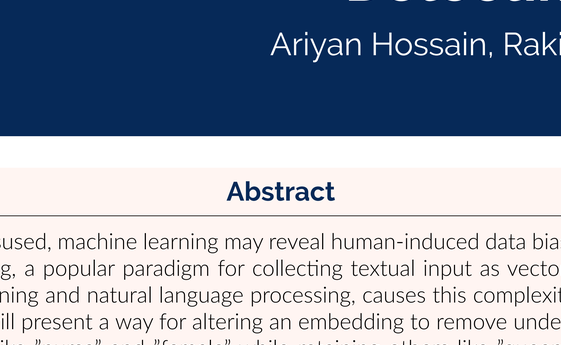


Figure 1. Gender Bias in Google Translate

Problem Statement

Gender bias present in word embeddings create a lot of issues in the downstream tasks. There is less research on bias present in recent models like BERT which produces contextual word. embeddings. It is a challenge to detect and mitigate gender bias from such models.

Research Objective

1. Analyze the existing bias for static and contextualized word embeddings.
2. We look at strategies for identifying bias in contextualized word embeddings.
3. We employ a BERT transformer model to find the existing bias.
4. We continue to focus on eliminating the bias that the results we obtained demonstrate.
5. We compare our model before and after debiasing.

Methodology

With the use of a multi-layer transformer architecture that includes both self-attention layers and feed-forward neural networks, BERT is able to process tokenized input text and generate contextualized word embeddings [1].

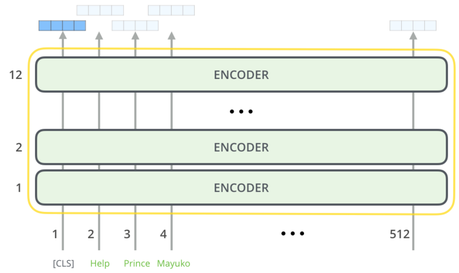


Figure 2. BERT architecture [1]

Pretrained on two unsupervised task:

- Masked Language Modeling:
- Next Sentence Prediction

For detecting bias, we gave sentences to BERT with masked token to predict the word. Experiment was repeated with many variations of BERT.

Models
bert-base-uncased
bert-large-uncased
distilbert-base-uncased
roberta-base
roberta-large
albert-base-v1
albert-large-v1

Table 1. Models used

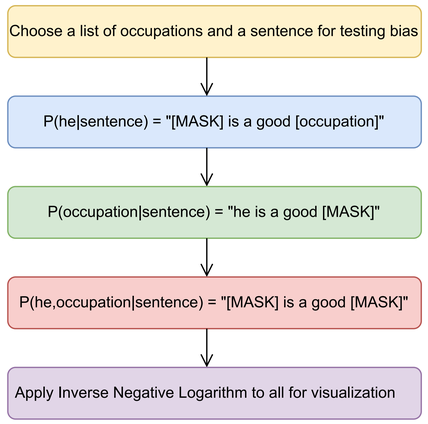


Figure 3. Mask Prediction Procedure

Dataset

For datasets, we used publicly available dataset for detecting gender bias.

- **WinoBias Dataset:** Rudinger et al. (2018)'s [2] dataset includes gender-neutral professions and gendered pronouns. We generated cosine similarity between profession terms using these statements. We tested gender bias in word embeddings using 60 professions from this dataset.
- **Pretrained on large dataset:** BERT is a pre-trained transformer model that has been trained on massive amounts of data. All BERT variants were trained using the same sets of data, which include the Wikidata set, which contains cleaned articles in all languages constructed from the Wikipedia dump, the Book Corpus, and a dataset of 11,038 unpublished books.
- **Synthetic Dataset:** We created a synthetic dataset using the pretrained model instead of giving it data. To explain, we feed the model a few phrase structures and it creates sentences to identify gender bias. These synthetic data—sentences, probability scores, normalized probability scores, and BERT model variation—were saved in a file.

Results and Analysis

- Masking the gendered pronoun:

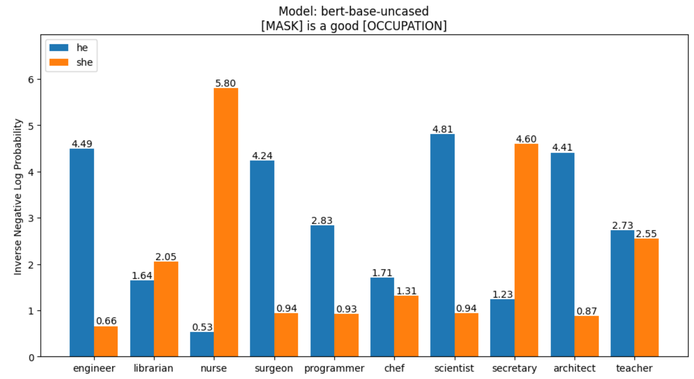


Figure 4. Predicting Gender Pronouns

- Masking the occupation:

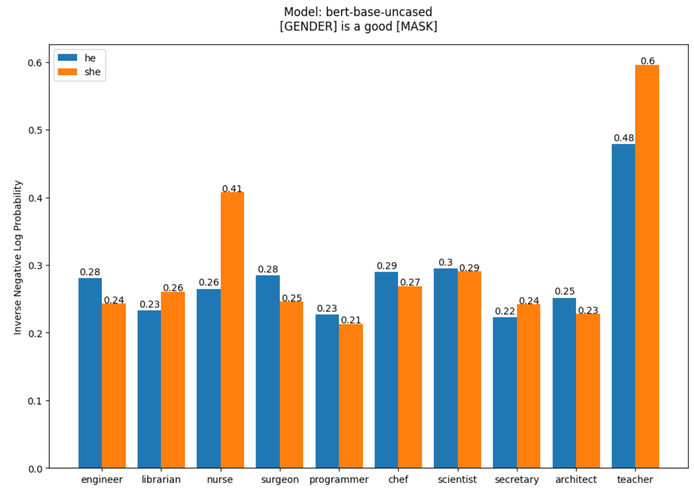


Figure 5. Predicting Occupations

- Masking both gendered pronouns and occupation:

Model	Input Sentence	Output Sentence
bert-base-uncased	[MASK] is a good [MASK]	he is a good teacher

Table 2. Predicting Gender Pronouns and Occupations

References

[1] Jay Alammar.
The illustrated transformer.
`https://jalammar.github.io/illustrated-transformer/`, 2018.

[2] Rachel Rudinger, Jason Naradowsky, Brian Leonard, and Benjamin Van Durme.
Gender bias in coreference resolution.
In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, New Orleans, Louisiana, June 2018. Association for Computational Linguistics.